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PHYSICS PTA - MODEL QUESTION PAPER -4
STANDARD XII

Maximum Marks : 70

PART – I

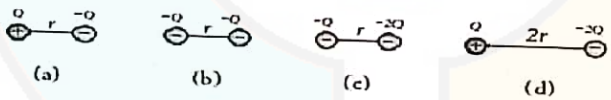
15× 1 =15

1. At infinity, the electrostatic potential is
   a) infinity b) maximum c) minimum **d) zero**
2. Gamma rays are used in the treatment of ____
   a) cancer b) polio c) AIDS d) tuberculosis
3. The phase difference between V_L *and* V_C in series RLC circuit.
   a) 2π b) $\frac{\pi}{2}$ c) $\frac{2\pi}{3}$ **d) π**
4. The internal resistance of a 2.1 V cell which gives a current of 0.2 A through a resistance of 10Ω is
   a) 0.2 Ω **b) 0.5 Ω** c) 0.8 Ω d) 1.0 Ω

   Hints: I = 0.2A, R = 10 Ω, V= IR=0.2 × 10 = 2 V; $r = \left(\frac{\xi - V}{V}\right) R = \left(\frac{2.1-2}{2}\right) 10 = 0.5\ \Omega$
5. In a tangent galvanometer of current 1A produces deflection of 30° the current required to produce a deflection of 60° is
   a) 3A b) 2A c) $\sqrt{2}$ A d) $\sqrt{3}$ A

   Hints : $I_1 = K \tan\theta_1$, $I_2 = K \tan\theta_2$; $\frac{I_1}{I_2} = \frac{\tan\theta_1}{\tan\theta_2}$; $I_2 = I_1 \times \frac{\tan\theta_1}{\tan\theta_2} = 1 \times \frac{\tan 60^\circ}{\tan 30^\circ} = \frac{\sqrt{3}}{\left(\frac{1}{\sqrt{3}}\right)} = 3A$
6. Dimension of resistance is
   a) $ML^2T^{-3}A^{-2}$ b) $ML^2T^{-1}A^{-1}$ c) $ML^2T^{-2}A^{-3}$ d) $ML^2T^{-1}A^{-2}$

   Hints: $H = I^2Rt$; $R = \frac{H}{I^2t} = \frac{ML^2T^{-2}}{A^2T} = ML^2T^{-3}A^{-2}$
7. Number of images produced by two parallel plane mirrors is
   a) infinity b) zero c) 3 d) 8
8. Rank of the electrostatic potential energies for the above system of charges in increasing order.

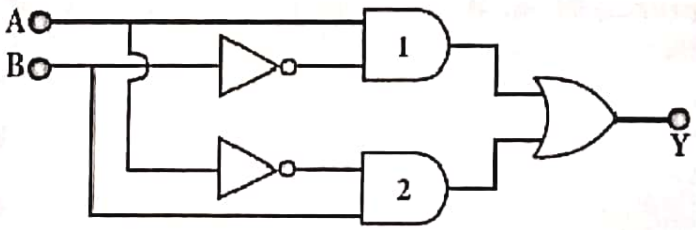


   a) 1= 4 < 2 < 3 b) 2 = 4 < 3 < 1 c) 2 = 3 < 1 < 4 d) 3 < 1 < 2 < 4

   Hints: $U_1 \propto \frac{-Q^2}{r}$; $U_2 \propto \frac{Q^2}{r}$; $U_3 \propto \frac{2Q^2}{r}$; $U_4 \propto \frac{-2Q^2}{2r} \propto \frac{-Q^2}{r}$
9. Relation between Half life and average life of radioactivity is
   a) $T_{1/2} = 1/\tau$ b) $T_{1/2} = 0.6931/\tau$ c) $T_{1/2} = \tau/2$ **d) $T_{1/2} = 0.6931\tau$**
10. In a Young's double slit experiment, the slit separation is doubled. To maintain the same fringe spacing on the screen, the screen-to-slit distance D must be changed to,
    a) 2D b) 3D c) $\sqrt{2}$D d) $\sqrt{2}$/D

    Hints: $\beta_1 = \frac{\lambda D}{d}$; $\beta_2 = \frac{\lambda D'}{2d}$; $\beta_1 = \beta_2$; $= \frac{\lambda D}{d} = \frac{\lambda D'}{2d}$; $D' = 2D$
11. Which one of the following is the natural nanomaterial.
    a) Peacock feather b) Peacock beak c) Grain of sand d) Skin of the whale
12. Blue colour LED is made up of
    a) SiC b) AlGaP c) GaAsP d) GaInN
13. What is the output Y in the above circuit, when all the three inputs A,B and C are 0?
    a) 0 **b) 1** c) 10 d) 11

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Hints: $X = A.B = 0 \cdot 0 = 0$; $Y = \overline{X.C} = \overline{0 \cdot 0} = \bar{0} = 1$

14. Momentum of electron having kinetic energy 2 eV is

a) 7.63×10^{-25} kg ms^{-1} b) 7.63×10^{-24} kg ms^{-1}

c) 6.63×10^{-25} kg ms^{-1} d) 6.63×10^{-24} kg ms^{-1}

15. The height of the transmitting and receiving antenna must be a multiple of ____

a) 2λ b) $\lambda/4$ c) $\lambda/2$ d) 4λ

Hints: $p = \sqrt{2mk} = \sqrt{2 \times 9.1 \times 10^{-31} \times 2 \times 1.6 \times 10^{-19}} = 7.636 \times 10^{-25}$kg m s^{-1}

PART – II

Answer any six questions . Question number 24 compulsory. $6 \times 2 = 12$

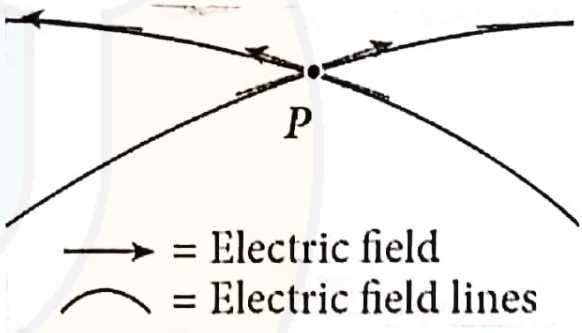
16. What is the principle of reversibility?

The principle of reversibility state that light will follow exactly the same path if its direction of travel is reversed.

17. The electric field lines never intersect. Justify.

Two electric field lines never intersect each other. If two lines cross at a point, then there will be two different electric field vectors at the same point, as shown in Figure

As a consequence, if some charge is placed in the intersection point, then it has to move in two different directions at the same time, which is physically impossible. Hence, electric field lines do not intersect.



18. In a transmitter connected in a common base configuration, $\alpha = 0.95$, $I_E = 1$ mA. Calculate the values of I_C and I_B.

$\alpha = \frac{I_C}{I_E}$; $I_C = \alpha I_E = 0.95 \times 1 \times 10^{-3} = 0.95$ mA

$I_E = I_B + I_C$; $I_B = I_E - I_C = 1 - 0.95 = 0.05$ mA

19. State Ampere's circuital law.

The line integral of magnetic field over a closed loop is μ_0 times net current enclosed by the loop. $\oint \vec{B} \cdot \vec{dl} = \mu_0 I_{enclosed}$

where $I_{enclosed}$ is the net current linked by the closed loop C.

20. Define temperature co-efficient of resistance.

The temperature coefficient of resistance is defined as the ratio of increase in resistance per degree rise in temperature to its resistance at T_o.

21. What is meant by Fraunhofer lines?

When the spectrum obtained from the Sun is examined, it consists of large number of dark lines (line absorption spectrum). These dark lines in the solar spectrum are known as Fraunhofer lines

22. What is meant by wattless current?

The current in an AC circuit is said to be Wattless current when the average power consumed in such circuit corresponds to zero.

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23. Half lives of two radioactive elements A and B are 20 minutes and 40 minutes respectively. Initially, the samples have equal number of nuclei. Calculate the ratio of decayed numbers of A and B nuclei after 80 minutes.

Remaining part of the element $= \frac{N}{N_o} = \left(\frac{1}{2}\right)^n$

Decayed part of the element $= 1 - \left(\frac{1}{2}\right)^n$

Where n = Number of Half life $= \frac{\text{Total life}}{\text{Half life}}$

Half life of A = 20 minutes; $n = \frac{80}{20} = 4$

Decayed part of the element $= 1 - \left(\frac{1}{2}\right)^4 = 1 - \frac{1}{16} = \frac{15}{16}$

Half life of A = 20 minutes; $n = \frac{80}{40} = 2$

Decayed part of the element $= 1 - \left(\frac{1}{2}\right)^2 = 1 - \frac{1}{4} = \frac{3}{4}$

Ratio of the decayed part of the elements A and B $= \frac{15}{16} : \frac{3}{4} = 60 : 48 = 5 : 4$

24. What should be the velocity of the electron so that its momentum equals that of 4000 Å wavelength photon.

$$\lambda_p = \frac{h}{p} = = \frac{h}{mv}$$

$$v = \frac{h}{m\lambda_p} = \frac{6.626 \times 10^{-34}}{9.11 \times 10^{-31} \times 4000 \times 10^{-10}} = 1818 \text{ ms}^{-1}$$

PART - III

Answer any six questions . Question number 33 compulsory. **$6 \times 3 = 18$**

25. Elaborate any two types of Robots with relevant examples.

Human Robot:

Certain robots are made to resemble humans in appearance and replicate the human activities like walking , lifting and sensing etc.

Industrial Robot:

There are six type industrial robots namely Cartesian, SCARA, cylindrical, Delta , Polar and vertically articulated. They are used in Arc Welding , Spot Welding, Material Handling, Machine Tending and other applications.

26. How will you induce an emf by changing the area enclosed by the coil?

Consider a conducting rod of length *l* moving with a velocity towards left on a rectangular metallic framework as shown in Figure.

The whole arrangement is placed in a uniform magnetic field whose magnetic lines are perpendicularly directed into the plane of the paper.

As the rod moves from AB to DC in a time dt, the area enclosed by the loop and hence the magnetic flux through the loop decreases.

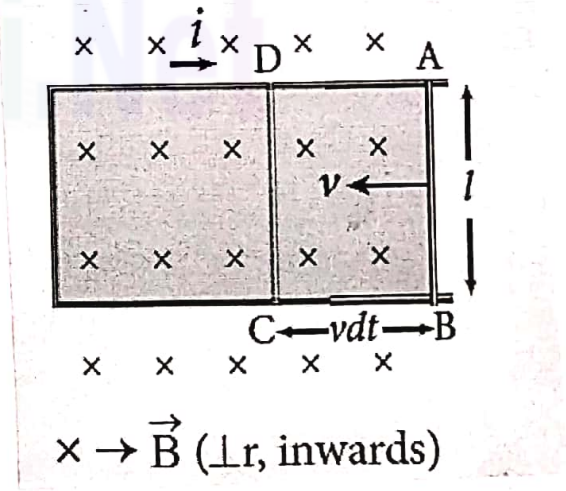
The change in magnetic flux in time dt is

$d\phi_B$ = B × change in area

= B × Area ABCD

= B*l*v dt since Area ABCD = *l*(vdt)

(or) $\frac{d\phi_B}{dt} = Blv$



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As a result of change in flux , an emf is generated in the loop .

The magnitude of the induced emf is

$$\varepsilon = \frac{d\phi_B}{dt}$$

$$\varepsilon = Blv$$

This emf is called **motional emf.**

The direction of induced current is found to be clockwise from Fleming's right hand rule .

27. Two cells each of 5V are connected in series across a 8Ω resistor and three parallel resistors of 4Ω, 6Ω and 12Ω. Draw a circuit diagram for the above arrangement. Calculate (i) the current drawn from the cell

(ii) current through each resistor

R_2,R_3 and R_4 are in parallel.

$$\frac{1}{R_p} = \frac{1}{R_2} + \frac{1}{R_3} + \frac{1}{R_4} = \frac{1}{4} + \frac{1}{6} + \frac{1}{12} = \frac{3+2+1}{12} = \frac{6}{12} = \frac{1}{2}$$

$$R_p = 2\Omega$$

R_1 and R_p are in series

$$R = R_1 + R_P = 8 + 2 = 10\ \Omega$$

i) Current drawn from the cell $I = \frac{V}{R} = \frac{10}{10} = 1$ A

Potential difference across 8 Ω resistor $V_1 = IR_1 = 1 \times 8 = 8$ V

Potential difference across 2 Ω resistor $V_2 = IR_2 = 1 \times 2 = 2$ V

ii) Current through the 8 Ω resistor $I_{R_1} = \frac{V_1}{R_1} = \frac{8}{8} = 1$A

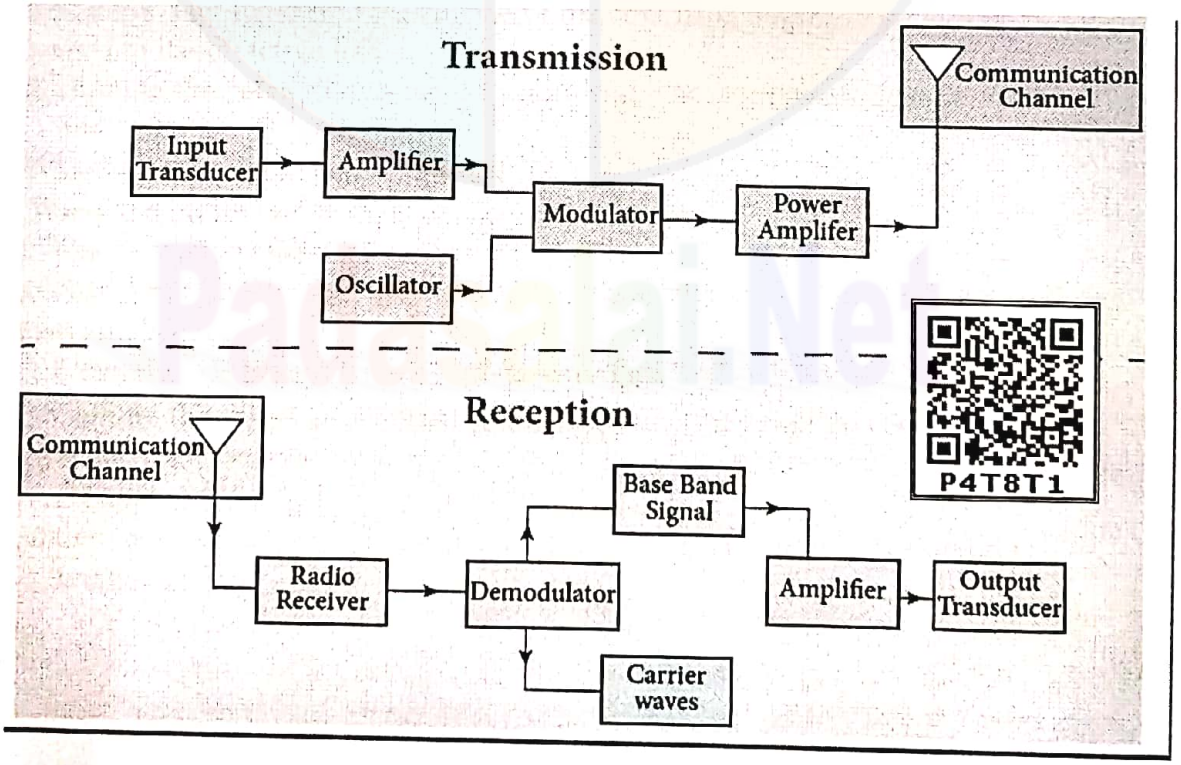
Current through the 4 Ω resistor $I_{R_2} = \frac{V_2}{R_2} = \frac{2}{4} = 0.5$A

Current through the 6 Ω resistor $I_{R_3} = \frac{V_2}{R_3} = \frac{2}{6} = \frac{1}{3} = 0.333$ A

Current through the 12 Ω resistor $I_{R_4} = \frac{V_2}{R_4} = \frac{2}{12} = \frac{1}{6} = 0.167$A

$\therefore I = 1$ A , $I_{R_1} = 1$A; $I_{R_2} = 0.5$A ; $I_{R_3} = 0.333$ A ; $I_{R_4} = 0.167$A

28. Elaborate on the basic elements of communication system with the necessary block diagram.



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29. Discuss the conversion of galvanometer into ammeter.

Ammeter is an instrument used to measure current flowing in the electrical circuit.

A galvanometer is converted into an ammeter by connecting a low resistance in parallel with the galvanometer. This low resistance is called shunt resistance S.

The scale is now calibrated in ampere and the range of ammeter depends on the values of the shunt resistance.

Let I be the current passing through the circuit as shown in Figure

When current I reaches the junction A, it divides into two components.

Let I_g be the current passing through the galvanometer of resistance R_g through a path AGE and the remaining current $(I - I_g)$ passes along the path ACDE through shunt resistance S.

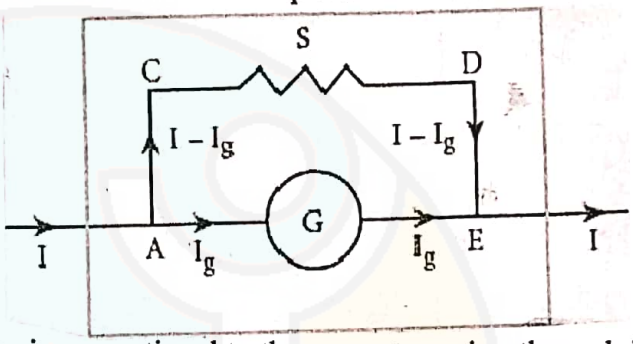
The potential difference across galvanometer is same as the potential difference across shunt resistance

$$V_{galvanometer} = V_{shunt}$$

$$I_g R_g = (I - I_g)\, S$$

$$S = \frac{I_g}{(I - I_g)} R_g$$

(or) $$I_g = \frac{S}{S + R_g} I \quad \Rightarrow I_g \propto I$$



Since, the deflection in the galvanometer is proportional to the current passing through it.

The effective resistance $\frac{1}{R_{eff}} = \frac{1}{R_g} + \frac{1}{S} \Rightarrow R_{eff} = \frac{R_g S}{R_g + S} = R_a$

The resistance offered by the ammeter is small. So, when we connect ammeter in series

For an ideal ammeter, the resistance must be equal to zero.

The percentage error in measuring a current through an ammeter is

$$\frac{\Delta I}{I} \times 100\% = \frac{I_{ideal} - I_{actual}}{I_{ideal}} \times 100\ \%$$

30. (i) State Brewster's law

Brewster's law states that the tangent of the polarising angle for a transparent medium is equal to its refractive index.

(ii) Calculate the refractive index of a material whose polarising angle is 60°

$n = \tan i_p = \tan 60° = 1.732$

31. (i) What is isotope? Give an example.

(ii) What is isotone? Give an example.

(iii) What is isobar? Give an example.

(i) Isotopes are atoms of the same element having same atomic number Z, but different mass number A.

Example: $^{1}_{1}H$ (hydrogen), $^{2}_{1}H$ (deuterium),and $^{3}_{1}H$ (tritium).

(ii) Isobars are the atoms of different elements having the same mass number A, but different atomic number Z.

Example : $^{40}_{17}S$; $^{40}_{18}Cl$; $^{40}_{19}Ar$; $^{40}_{19}K$; $^{40}_{20}Ca$

(iii) Isotones are the atoms of different elements having same number of neutrons.

Example : $^{12}_{5}B$; $^{13}_{6}C$

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32. Calculate the range of the variable capacitor that is to be used in a tuned-collector oscillator which has a fixed inductance of 150μH. The frequency band is from 500 kHz to 1500 kHz.

$f = \frac{1}{2\pi\sqrt{LC}}$; $f^2 = \frac{1}{4\pi^2 LC}$; $C = \frac{1}{4\pi^2 L f^2}$

$L = 150\ \mu H = 150 \times 10^{-6}$ H; $f_1 = 500$ kHz $= 500 \times 10^3$ Hz

$$C_1 = \frac{1}{4\pi^2 L f_1^2} = \frac{1}{4\pi^2 \times 150\times10^{-6} \times (500\times10^3)^2} = 676\text{pF}$$

$$C_2 = \frac{1}{4\pi^2 L f_2^2} = \frac{1}{4\pi^2 \times 150\times10^{-6} \times (1500\times10^3)^2} = 75\text{pF}$$

Range of the variable capacitor = 75 pF to 676pF

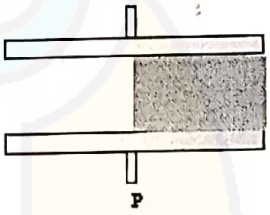
33. Capacitors P and Q have identical cross sectional areas A and separation d. The space between the capacitors is filled with a dielectric of dielectric constant ε_r as shown in the figure. Calculate the capacitance of capacitors P and Q.

i) C_1,C_2 are in parallel

$$C_1 = \frac{\varepsilon_0 \frac{A}{2}}{d} \qquad C_2 = \frac{\varepsilon_0 \varepsilon_r \frac{A}{2}}{d}$$

$$C_p = C_1 + C_2 = \frac{\varepsilon_0 \frac{A}{2}}{d} + \frac{\varepsilon_0 \varepsilon_r \frac{A}{2}}{d} = \frac{\varepsilon_0 A}{2d} + \frac{\varepsilon_0 \varepsilon_r A}{2d}$$

$$C_p = \frac{\varepsilon_0 A}{2d}[1 + \varepsilon_r]$$



ii) C_1,C_2 are in series

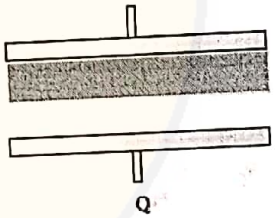
$$C_1 = \frac{\varepsilon_0 \varepsilon_r A}{\frac{d}{2}} = \frac{2\varepsilon_0 \varepsilon_r A}{d}$$

$$C_2 = \frac{\varepsilon_0 A}{\frac{d}{2}} = \frac{2\varepsilon_0 A}{d}$$

$$\frac{1}{C_Q} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{d}{2\varepsilon_0\varepsilon_r A} + \frac{d}{2\varepsilon_0 A} = \frac{d}{2\varepsilon_0 A}\left[\frac{1}{\varepsilon_r} + 1\right]$$

$$= \frac{d}{2\varepsilon_0 A}\left(\frac{\varepsilon_r + 1}{\varepsilon_r}\right)$$

$$C_Q = \frac{2\varepsilon_0 A}{d}\left(\frac{\varepsilon_r}{\varepsilon_r + 1}\right)$$

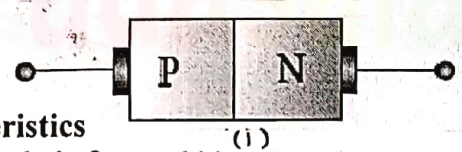


PART - IV

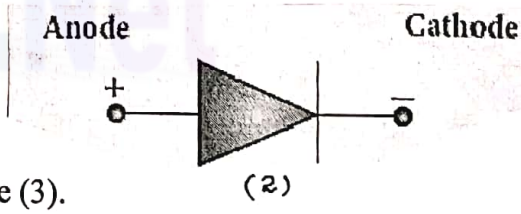
Answer all the questions $5 \times 5 = 25$

34. a) Explain the formation of PN junction diode. Discuss its V–I characteristics.

A p-n junction diode is formed when a p-type semiconductor is fused with a n-type semiconductor. It is a device with single p-n junction as shown in Figure (1).
The circuit symbol is shown in Figure (2).



(1)



(2)

Forward characteristics

The p-n junction diode is forward biased as shown in Figure (3).
An external resistance (R) is used to limit the flow of current through the diode.
The voltage across the diode is varied by varying the biasing voltage across the dc power supply.
The forward bias voltage and the corresponding forward bias current are noted.
A graph is plotted by taking the forward bias voltage (V) along the X-axis and the current (I) through the diode along the Y-axis. This graph is called the **forward V-I characteristics** of the p-n junction diode and is shown in Figure (4)

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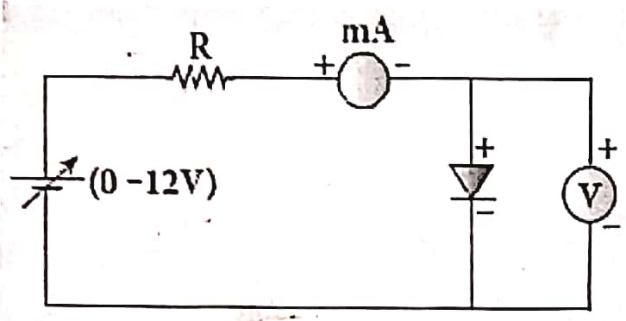


Fig: (3)

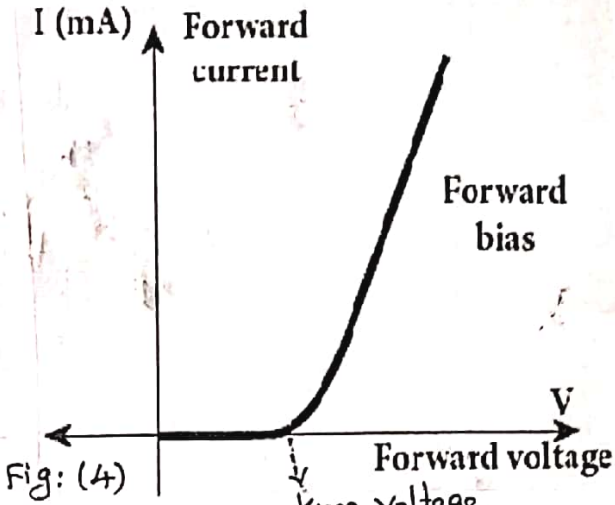


Fig: (4)

(i) At room temperature, a potential difference equal to the barrier potential is required before a reasonable forward current starts flowing across the diode. This voltage is known as **threshold voltage or cut-in voltage or knee voltage .**
It is approximately 0.3 V for Germanium and 0.7 V for Silicon.

(ii) The graph clearly infers that the current flow is not linear and is exponential. Hence it does not obey Ohm's law.

(iii) The forward resistance $r_f = \frac{\Delta V}{\Delta I}$

.(iv) Thus the diode behaves as a conductor when it is forward biased.

Reverse characteristics

The circuit to study the reverse characteristics is shown in Figure 5.
In the reverse bias, the p-region of the diode is connected to the negative terminal and n-region to the positive terminal of the dc power supply.
A graph is drawn between the reverse bias voltage and the current across the junction, which is called the reverse characteristics of a p-n junction diode.
It is shown in Figure (6).

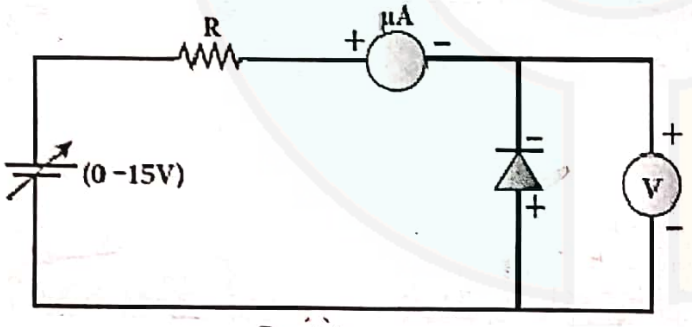


Fig: (5)

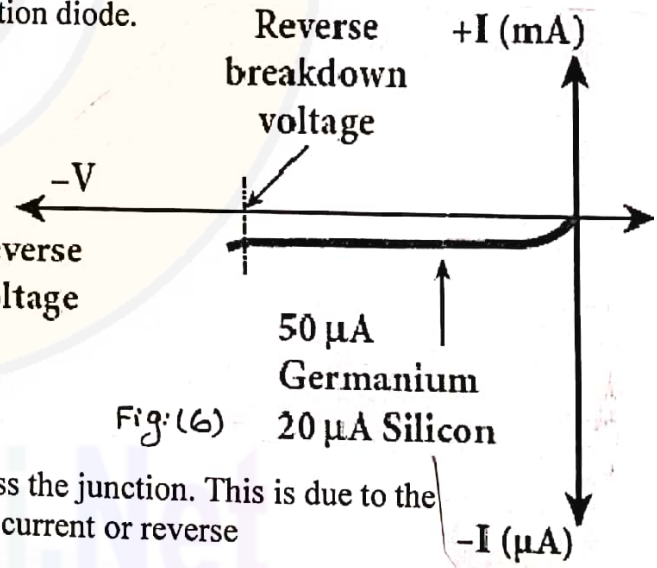


Fig: (6)

Under this bias, a very small current in μA, flows across the junction. This is due to the flow of the minority charge carriers called the leakage current or reverse saturation current.

(OR)

b). i) Derive an expression for de Broglie wavelength of electrons

An electron of mass *m* is accelerated through a potential difference of V volt.
The kinetic energy acquired by the electron is given by

$$\frac{1}{2}mv^2 = eV$$

Therefore, the speed v of the electron is

$$v = \sqrt{\frac{2eV}{m}}$$

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Hence, the de Broglie wavelength of the electron is

$$\lambda = \frac{h}{mv} = \frac{h}{\sqrt{2emV}}$$

Substituting the known values in the above equation, we get

$$\lambda = \frac{6.626 \times 10^{-34}}{\sqrt{2V \times 1.6 \times 10^{-11} \times 9 \times 10^{-31}}}$$

$$= \frac{12.27 \times 10^{-10}}{\sqrt{V}} = \frac{12.27}{\sqrt{V}} \text{Å}$$

For example, if an electron is accelerated through a potential difference of $100V$, then its de Broglie wavelength is 1.227 Å

Since the kinetic energy of the electron,K = eV, then the de Broglie wavelength associated with electron can be also written as

$$\lambda = \frac{h}{\sqrt{2mK}}$$

ii) Calculate the de Broglie wavelength of a proton whose kinetic energy is equal to 81.9×10^{-15} J. (Given: mass of proton is 1836 times that of electron

$$\lambda_p = \frac{h}{\sqrt{2m_pK}} = \frac{h}{\sqrt{2\times(1836m_e)K}}$$

$$= \frac{6.626\times10^{-34}}{\sqrt{2\times1836\times9.11\times10^{-31}\times81.9\times10^{-15}}}$$

$$= \frac{6.626\times10^{-14}}{\sqrt{2.74}} = \frac{6.626\times10^{-14}}{1.655}$$

$$= 4 \times 10^{-14} \text{ m}$$

35. a) What do you know about GPS? Write a few applications of GPS.

GPS stands for Global Positioning System.

It is a global navigation satellite system that offers geolocation and time information to a GPS receiver anywhere on or near the Earth.

GPS system works with the assistance of a satellite network.

Each of these satellites broadcasts a precise signal like an ordinary radio signal. These signals that convey the location data are received by a low-cost aerial which is then translated by the GPS software.

The software is able to recognize the satellite, its location, and the time taken by the signals to travel from each satellite.

The software then processes the data it accepts from each satellite to estimate the location of the receiver.

Applications

Global positioning system is highly useful many fields such as

(i) fleet vehicle management (for tracking cars, trucks and buses),

(ii) wildlife management (for counting of wild animals) and

(iii) engineering (for making tunnels, bridges etc).

(OR)

b) Write down any ten properties of electromagnetic waves

(i) Electromagnetic waves are produced by any accelerated charge.

(ii) They do not require any medium for propagation. So electromagnetic waves are non-mechanical wave.

(iii) They are transverse in nature, (i.e) the oscillating electric field vector, oscillation magnetic field vector and direction of propagation are mutually perpendicular to each other

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(iv) They travel with speed of light in vacuum or free space and it is given by,

$C = \frac{1}{\sqrt{\varepsilon_o \mu_o}} = 3\times 10^8\ ms^{-1}$

(v) In a medium with permittivity 'ε' and permeability 'μ', the speed of electromagnetic wave is less than speed in free space or vacuum. (i.e.) $v<c$

Hence, refractive index of the medium is, $\mu = \frac{c}{v} = \frac{\frac{1}{\sqrt{\varepsilon_o \mu_o}}}{\frac{1}{\sqrt{\varepsilon \mu}}} = \sqrt{\varepsilon_r \mu_r}$

(vi) They are not deflected by electric or magnetic field.

(vii) They show interference, diffraction and polarization.

(viii) The energy density (energy per unit volume) associated with and electro magnetic wave propagating in free space is

u = Energy density in an electric field + Energy density in an magnetic field

$= \frac{1}{2}\varepsilon_o E^2 + \frac{1}{2\mu_o} B^2$

(ix) The average energy density for electromagnetic wave is $\langle u \rangle = \frac{1}{2}\varepsilon_o E^2 = \frac{1}{2\mu_o} B^2$

(x) The energy crossing per unit area per unit time and perpendicular to the direction of propagation of electromagnetic wave is called the intensity.

(xi) They carry energy and momentum. The force exerted by an electromagnetic surface is called ***radiation pressure***.

(xii) If the electromagnetic wave incident on a material surface is completely absorbed, then the energy delivered is 'U' and the momentum imparted on the suface is $p = \frac{U}{C}$

(xiii) If the incident electro magnetic wave of energy 'U' is totally reflected from the surface, then the momentum delivered to the surface is , $\Delta p = \frac{U}{C} - \left(-\frac{U}{C}\right) = 2\frac{U}{C}$

(xiv) The rate of flow of energy crossing a unit area is known as pointing vector for electromagnetic waves.

36. a) Derive the expression for the radius of nth orbit of a hydrogen atom using Bohr atom model

Consider an atom which contains the nucleus at rest and an electron revolving around the nucleus in a circular orbit of radius r_n as shown in Figure.

Nucleus is made up of protons and neutrons. Since proton is positively charged and neutron is electrically neutral, the charge of a nucleus is purely the total charge of protons.

Let Z be the atomic number of the atom then $+Ze$ is the charge of the nucleus.

Let $-e$ be the charge of the electron.

From Coulomb's law, the force of attraction between the nucleus and the electron is

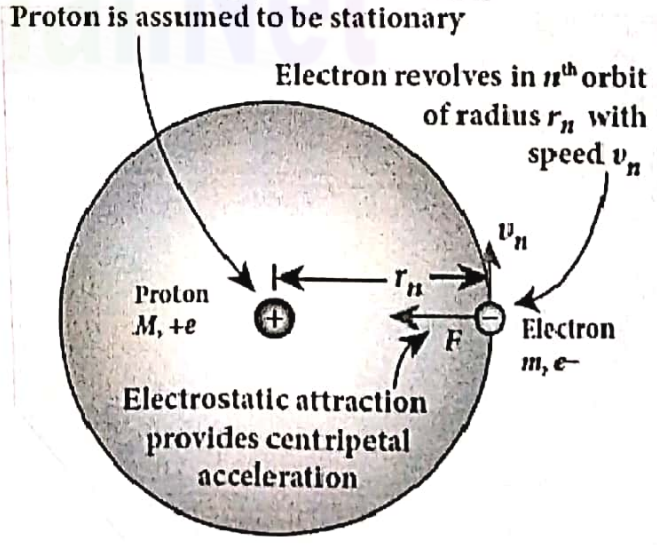
$$\vec{F}_{coloumb} = \frac{1}{4\pi\varepsilon_o}\frac{(+Ze)(-e)}{r_n^2}\hat{r}$$

$$= -\frac{1}{4\pi\varepsilon_o}\frac{Ze^2}{r_n^2}\hat{r}$$

This force provides necessary centripetal force

$$\vec{F}_{centripetal} = \frac{mv_n^2}{r_n}\hat{r}$$

where m be the mass of the electron and v_n be the velocity in a circular orbit.



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Therefore,

$$|\vec{F}_{coloumb}| = |\vec{F}_{centripetal}| = \frac{1}{4\pi\varepsilon_o}\frac{Ze^2}{r_n^2} = \frac{mv_n^2}{r_n}$$

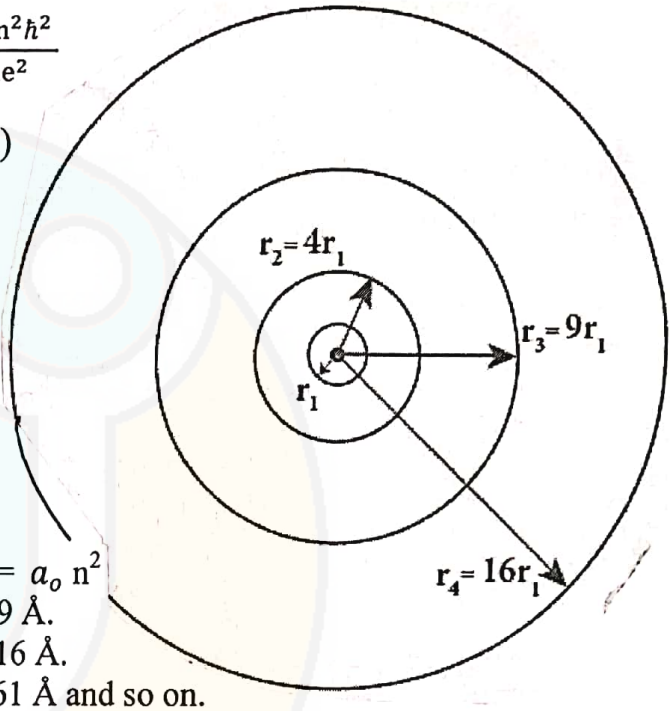
$$r_n = \frac{4\pi\varepsilon_o(mv_n r_n)^2}{Zme^2}$$

From Bohr's assumption, the angular momentum quantization condition, $mv_n r_n = l_n = n\hbar$

$$\therefore r_n = \frac{4\pi\varepsilon_o(mv_n r_n)^2}{Zme^2} = \frac{4\pi\varepsilon_o(n\hbar)^2}{Zme^2} = \frac{4\pi\varepsilon_o n^2\hbar^2}{Zme^2}$$

$$r_n = \left(\frac{\varepsilon_o h^2}{\pi me^2}\right)\frac{n^2}{Z} \qquad \left(\because \hbar = \frac{h}{2\pi}\right)$$

Where $n \in \mathbb{N}$, ε_o,h,e and π are constants.



Therefore, the radius of the orbit becomes $r_n = a_o \frac{n^2}{Z}$

where $a_o = \frac{\varepsilon_o h^2}{\pi me^2} = 0.529$ Å

This is known as Bohr radius which is the smallest radius of the orbit in an atom.

Bohr radius is also used as unit of length called Bohr.

1 Bohr = 0.53 Å.

For hydrogen atom (Z = 1), the radius of nth orbit is $r_n = a_o n^2$

For the first orbit (ground state), $r_1 = a_0 = 0.529$ Å.

For the second orbit (first excited state), $r_2 = 4a_0 = 2.116$ Å.

For the third orbit (second excited state) $r_3 = 9a_0 = 4.761$ Å and so on.

Thus the radius of the orbit from centre increases with *n*, that is, $r_n \propto n^2$ as shown in Figure

(OR)

b) Obtain the equation for resolving power of microscope

The diagram related to the calculation of resolution of microscope is illustrated in Figure. A microscope is used to see the details of the object under observation. The ability of microscope depends not only in magnifying the object but also in resolving two points on the object separated by a small distance d_{min}. Smaller the value of d_{min} better will be the resolving power of the microscope.

The radius of central maxima is already derived as,

$$r_o = \frac{1.22\lambda v}{a}$$

In the place of focal length *f* we have the image distance *v*.

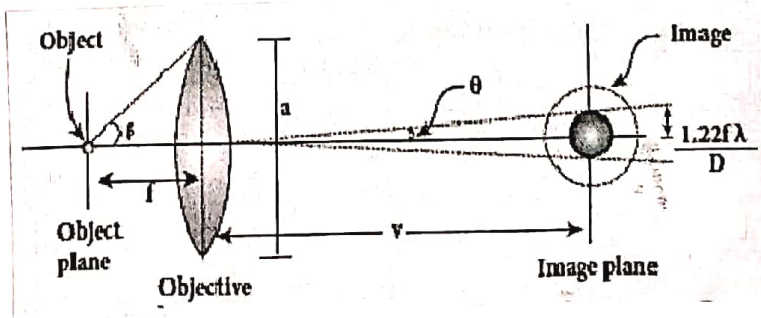
If the difference between the two points on the object to be resolved is d_{min},

magnification $m = \frac{r_o}{d_{min}}$

$$d_{min} = \frac{r_o}{m}$$

$$= \frac{1.22\lambda v}{am} = \frac{1.22\lambda v}{a\left(\frac{v}{u}\right)}$$

$$= \frac{1.22\lambda u}{a} \qquad [\because m = v/u]$$



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$$d_{min} = \frac{1.22\lambda f}{a} \quad [u \approx f]$$

On the object side,

$$2\tan\beta = 2\sin\beta = \frac{a}{f} \quad \left[\tan\beta = \frac{\left(\frac{a}{2}\right)}{f} \quad \therefore a = f2\sin\beta\right]$$

$$d_{min} = \frac{1.22\lambda}{2\sin\beta}$$

To further reduce the value of d_{min} the optical path of the light is increased by immersing the objective of the microscope in to a bath containing oil of refractive index n.

$$d_{min} = \frac{1.22\lambda}{2n\sin\beta}$$

Such an objective is called oil immersed objective. **The term nsinβ is called numerical aperture NA.**

$$d_{min} = \frac{1.22\lambda}{2\,(NA)}$$

37. a) Obtain a relation for the magnetic induction at a point along the axis of a circular coil carrying current .

Consider a current carrying circular loop of radius R and let I be the current flowing through the wire in the direction as shown in Figure. The magnetic field at a point P on the axis of the circular coil at a distance z from its center of the coil O. It is computed by taking two diametrically opposite line elements of the coil each of length at C and D. Let be the vector joining the current element (I) at C to the point P.

$$PC = PD = r = \sqrt{R^2 + Z^2}$$

$$\angle CPO = \angle DPO = (90 - \theta)$$

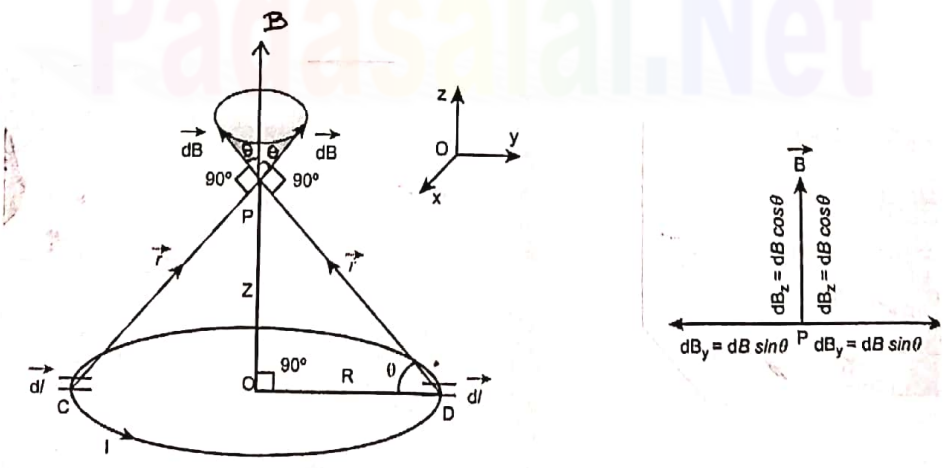
According to Biot-Savart's law, the magnetic field at P due to the current element $Id\vec{\ell}$ is

$$d\vec{B} = \frac{\mu_o}{4\pi}\frac{I\,d\vec{\ell}\times\hat{r}}{r^2}$$

The magnitude of magnetic field due to current element $Id\vec{\ell}$ at C and D are equal because of equal distance from the coil.

The magnetic field $d\vec{B}$ due to each current element $Id\vec{\ell}$ is resolved into two components; dB sin θ along y - direction and dB cos θ along z - direction. Horizontal components of each current element cancels out while the vertical components (dB cos θ $\hat{k}$) alone contribute to total magnetic field at the point P.

If we integrate around the loop, $d\vec{B}$ sweeps out a cone as shown in Figure,



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The net magnetic field at point P is

$\vec{B} = \int d\vec{B} = \int dB \cos\theta\, \hat{k}$

$\vec{B} = \frac{\mu_0 I}{4\pi} \int \frac{d\ell}{r^2} \cos\theta\, \hat{k} = \frac{\mu_0 I}{4\pi} \frac{1}{r^2} \cos\theta\, \hat{k} \int d\vec{\ell}$

using Pythagorous theorem $r^2 = R^2 + Z^2$ and $\cos\theta = \frac{R}{(R^2+Z^2)^{\frac{1}{2}}}$

and integrating line element from 0 to 2πR,

$\vec{B} = \frac{\mu_0 I}{4\pi} \frac{1}{r^2} \cos\theta\, \hat{k} \int_0^{2\pi R} d\vec{\ell} = \frac{\mu_0 I}{4\pi} \frac{1}{(R^2+Z^2)} \frac{R}{(R^2+Z^2)^{\frac{1}{2}}} \hat{k}\,(2\pi R)$

we get $\vec{B} = \frac{\mu_0 I}{2} \frac{R^2}{(R^2+Z^2)^{\frac{3}{2}}} \hat{k}$

Note that the magnetic field points along the direction from the point O to P.
Suppose if the current flows in clockwise direction, then magnetic field points in the direction from the point P to O.

(OR)

b) Derive an expression for electrostatic potential due to an electric dipole

Consider two equal and opposite charges separated by a small distance 2a as shown in Figure . The point P is located at a distance r from the midpoint of the dipole.
Let θ be the angle between the line OP and dipole axis AB.
Let r_1 be the distance of point P from +q and r_2 be the distance of point P from q.

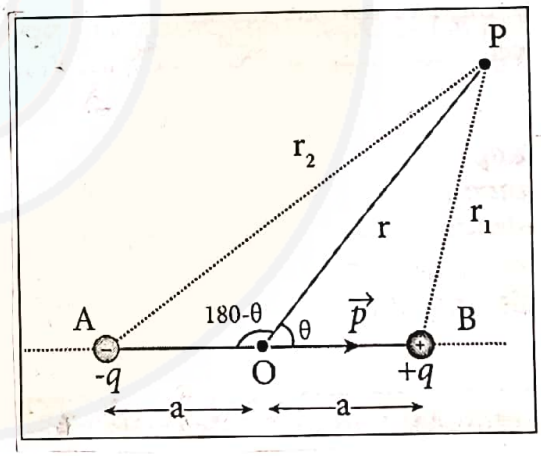
Potential at P due to charge +q = $\frac{1}{4\pi\varepsilon_0} \frac{q}{r_1}$

Potential at P due to charge –q= - $\frac{1}{4\pi\varepsilon_0} \frac{q}{r_2}$

Total potential at the point P,

$$V = \frac{1}{4\pi\varepsilon_0} q \left(\frac{1}{r_1} - \frac{q}{r_2} \right) \text{.....(1)}$$

Suppose if the point P is far away from the dipole, such that r>>a, then equation (1) can be expressed in terms of r.



By the cosine law for triangle BOP, $r_1^2 = r^2 + a^2 - 2ra\cos\theta$ $r_1^2 = r^2\left(1 + \frac{a^2}{r^2} - \frac{2a\cos\theta}{r}\right)$ Since the point P is very far from dipole, then r>>a. As a result the term $\frac{a^2}{r^2}$ is very small and can be neglected. Therefore $r_1^2 = r^2\left(1 - \frac{2a\cos\theta}{r}\right)$	Similarly applying the cosine law for triangle AOP, $r_2^2 = r^2 + a^2 - 2ra\cos(180 - \theta)$ $\cos(180 - \theta) = -\cos\theta$ $r_2^2 = r^2 + a^2 + 2ra\cos\theta$ Neglecting the term $\frac{a^2}{r^2}$(because r >>a) $r_2^2 = r^2\left(1 + \frac{2a\cos\theta}{r}\right)$

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(or) $r_1 = r\left(1 - \frac{2a\cos\theta}{r}\right)^{\frac{1}{2}}$ $\frac{1}{r_1} = \frac{1}{r}\left(1 - \frac{2a\cos\theta}{r}\right)^{-\frac{1}{2}}$ Since, $\frac{a}{r} << 1$, we can use binomial theorem and retain the terms up to first order $\frac{1}{r_1} = \frac{1}{r}\left(1 + \frac{a\cos\theta}{r}\right)$ ---------(2)	(or) $r_2 = r\left(1 + \frac{2a\cos\theta}{r}\right)^{\frac{1}{2}}$ $\frac{1}{r_2} = \frac{1}{r}\left(1 + \frac{2a\cos\theta}{r}\right)^{-\frac{1}{2}}$ Using Binomial theorem, we get $\frac{1}{r_2} = \frac{1}{r}\left(1 - \frac{a\cos\theta}{r}\right)$ ---------(3)

Substituting equation (2) and (3) in equation (1),

$$V = \frac{1}{4\pi\varepsilon_0} q \left\{\frac{1}{r}\left(1 + \frac{a\cos\theta}{r}\right) - \frac{1}{r}\left(1 - \frac{a\cos\theta}{r}\right)\right\}$$

$$V = \frac{1}{4\pi\varepsilon_0} q \left\{\frac{1}{r}\left(1 + \frac{a\cos\theta}{r} - 1 + \frac{a\cos\theta}{r}\right)\right\}$$

$$V = \frac{1}{4\pi\varepsilon_0} \frac{2aq}{r^2}\cos\theta$$

But the electric dipole moment p = 2qa and we get,

$$V = \frac{1}{4\pi\varepsilon_0}\left(\frac{p\cos\theta}{r^2}\right) \text{ .....(4)}$$

Now we can write $p\cos\theta = \vec{p} \cdot \hat{r}$ where $\hat{r}$ is the unit vector from the point O to point P. Hence the electric potential at a point P due to an electric dipole is given by

$$V = \frac{1}{4\pi\varepsilon_0}\frac{\vec{p}\cdot\hat{r}}{r^2} \qquad (r>>a) \quad \text{-----(5)}$$

Equation (5) is valid for distances very large compared to the size of the dipole. But for a point dipole, the equation (5) is valid for any distance.

Case (i)	Case (ii)	Case (iii)
If the point P lies on the axial line of the dipole on the side of +q, then $\theta = 0$. $V = \frac{1}{4\pi\varepsilon_0}\frac{\vec{p}}{r^2}$	If the point P lies on the axial line of the dipole on the side of –q, then $\theta = 180°$. $V = -\frac{1}{4\pi\varepsilon_0}\frac{\vec{p}}{r^2}$	If the point P lies on the equatorial line of the dipole, then $\theta = 90°$. Hence $V = 0$

38 a) Derive an expression for phase angle between the applied voltage and current in a series RLC circuit

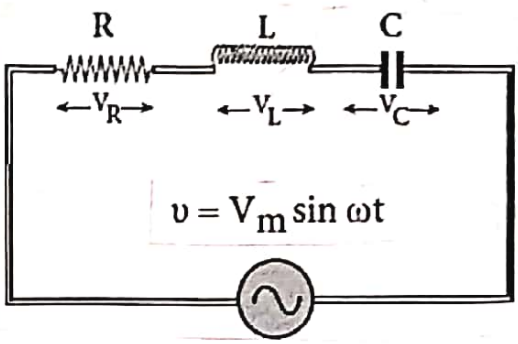
Consider a circuit containing a resistor of resistance R, a inductor of inductance L and a capacitor of capacitance C connected across an alternating voltage source (Figure).

The applied alternating voltage is given by the equation.

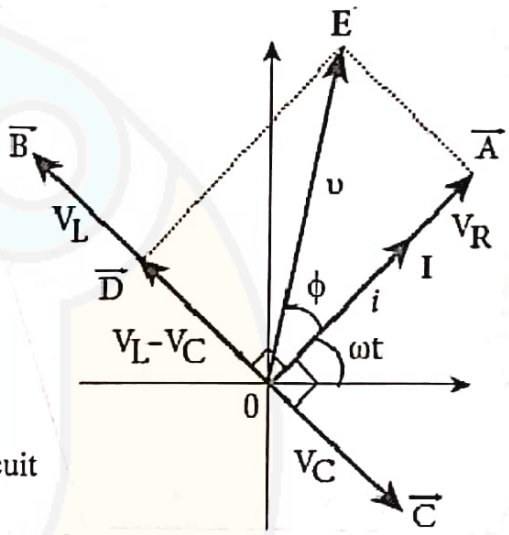
$$v = V_m \sin\omega t \text{ .....(1)}$$

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Let i be the resulting circuit current in the circuit at that instant. As a result, the voltage is developed across R, L and C.



We know that
voltage across R (V_R) is in phase with i,
voltage across L (V_L) leads i by $\pi/2$ and
voltage across C (V_C) lags i by $\pi/2$.
The phasor diagram is drawn with current as the reference phasor.
The current i is represented by the phasor $\overrightarrow{OI}$,
V_R by $\overrightarrow{OA}$, V_L by $\overrightarrow{OB}$ and V_C by $\overrightarrow{OC}$ as shown in Figure
The net voltage drop across L-C combination is $V_L - V_C$ which is represented by a phasor $\overrightarrow{OD}$
By parallelogram law, the diagonal $\overrightarrow{OE}$ gives the resultant voltage υ of V_R and $(V_L - V_C)$ and its length OE is equal to V_m. Therefore,



$$V_m^2 = V_R^2 + (V_L - V_C)^2$$

The effective voltage(V_m)

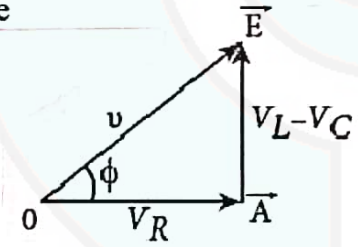
$$V_m = \sqrt{(I_m R)^2 + (I_m X_L - I_m X_C)^2}$$

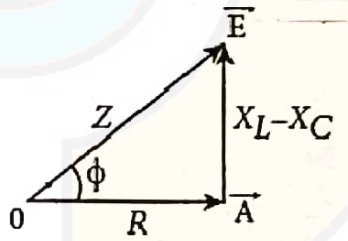
$$= I_m \sqrt{R^2 + (X_L - X_C)^2}$$

$$\text{(or) } I_m = \frac{V_m}{\sqrt{R^2 + (X_L - X_C)^2}}$$

$$\text{(or) } I_m = \frac{V_m}{Z}$$

Where $Z = \sqrt{R^2 + (X_L - X_C)^2}$ = Impedance of the circuit
The voltage triangle and impedance triangle are given in the Figure





From phasor diagram, the phase angle between υ and i is found out from the following relation

$$\tan \phi = \left(\frac{V_L - V_C}{R}\right) = \left(\frac{I_m X_L - I_m X_C}{I_m R}\right) = \left(\frac{X_L - X_C}{R}\right)$$

Special cases

(i) If $X_L > X_C$, $(X_L - X_C)$ is positive and phase angle ϕ is also positive. It means that the applied voltage leads the current by ϕ (or current lags behind voltage by ϕ). The circuit is inductive.

$$\therefore v = V_m \sin\omega t \ , \ i = I_m \sin(\omega t - \phi)$$

(ii) If $X_L < X_C$, $(X_L - X_C)$ is negative and ϕ is also negative. Therefore current leads voltage by ϕ and the circuit is capacitive.

$$\therefore v = V_m \sin\omega t \ , \ i = I_m \sin(\omega t + \phi)$$

(iii) If $X_L = X_C$, ϕ is zero. Therefore current and voltage are in the same phase and the circuit is resistive.

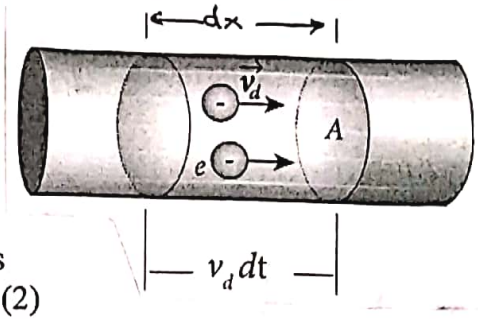
$$\therefore v = V_m \sin\omega t \ , \ i = I_m \sin\omega t$$

(OR)

b) i) Describe the microscopic model of current and obtain microscopic form of Ohm's law

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Consider a conductor with area of cross section A and an electric field $\vec{E}$ applied from right to left.



Number of electrons per unit volume in the conductor = n
drift velocity of the electrons = v_d
charge of the electrons = e
Let "dx' be the distance travelled by the electrons in time 'dt',

Drift velocity = $v_d = \frac{dx}{dt}$; $dx = v_d\, dt$ .....(1)

the number of electrons available in the volume of length dx is

$= A\, dx \times n = A\, v_d\, dt \times n$ .....(2)

Total charge in volume element

dQ = (charge) × (number of electrons per unit volume)

$dQ = (e)\,(A\, v_d\, dt)\, n$

Current = $I = \frac{dQ}{dt} = \frac{n\, e\, A\, v_d\, dt}{dt}$

$I = n\, e\, A\, v_d$ .....(3)

Current Density

The current density (J)is defined as the current per unit area of cross section of the conductor.

$J = \frac{I}{A}$ .....(4)

The S.I unit of current density is A m^{-2}

$J = \frac{neAv_d}{A}$

$J = nev_d$.....(5)

The above expression is valid only when the direction of the current is perpendicular to the area A.

We can write in vector form $\vec{J} = ne\vec{v}_d$.....(6)

Substitute $\vec{v}_d = -\frac{e\tau}{m}\vec{E}$ in (6)

$\vec{J} = ne\,(-\frac{e\tau}{m}\vec{E}) = -\frac{ne^2\tau}{m}\vec{E}$

Where $\frac{ne^2\tau}{m} = \sigma$ = conductivity

$\vec{J} = -\sigma\vec{E}$

But conventionally, we take the direction of (conventional) current density as the direction of electric field. So the above equation becomes

$\vec{J} = \sigma\vec{E}$

This is called microscopic form of ohm's law

ii) A copper wire of 10^{-6} m^2 area of cross section, carries a current of 2A. If the number of electrons per cubic meter is 8 x 10^{28}, calculate the current density and average drift velocity

$J = \frac{I}{A} = \frac{2}{10^{-6}} = 2 \times 10^6\ Am^{-2}$

$I = nAv_de$

$v_d = \frac{I}{nAe}$

$= \frac{2}{8\times10^{28}\times10^6\times1.6\times10^{-19}} = \frac{2}{8\times16\times10^2}$

$= 1.5625 \times 10^{-4}\ ms^{-1}$

- T. JOHN PETER -

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